

Lecture Note 3: Useful Inequalities and Large Sample Theory

14.380: Statistical Methods in Economics, Fall 2026

1 Useful Inequalities

We now introduce some useful inequalities that will appear throughout econometrics and other areas of economics.

Theorem 1 (Jensen's Inequality). *Let $h(\cdot)$ be a convex function and X be a random variable. Then,*

$$\mathbb{E}[h(X)] \geq h(\mathbb{E}[X]).$$

Proof. Recall that if $h(\cdot)$ is a convex function, then $\forall x_0$ in its domain, there exists a line through $(x_0, h(x_0))$ such that $h(x)$ never falls below the line. That is, there exists some constant a such that

$$h(x) \geq h(x_0) + a(x - x_0) \quad \forall x$$

Set $x_0 = \mathbb{E}[X]$. Evaluating this inequality at the random variable X , it follows that

$$h(X) \geq h(\mathbb{E}[X]) + a(X - \mathbb{E}[X]).$$

Taking expectations, we have that

$$\mathbb{E}[h(X)] \geq h(\mathbb{E}[X]).$$

□

Notice that if $h(\cdot)$ is concave, then the opposite inequality holds; that is, $\mathbb{E}[h(X)] \leq h(\mathbb{E}[X])$.

The next inequality (Markov's Inequality) provides a bound on the tail behavior of a random variable as a function of its expectation.

Theorem 2 (Markov's Inequality). *Suppose X is a random variable with $X \geq 0$ and $\mathbb{E}[X] < \infty$. Then, for any $M > 0$,*

$$P(X \geq M) \leq \frac{\mathbb{E}[X]}{M}.$$

Proof. Since X is non-negative,

$$\begin{aligned} \mathbb{E}[X] &= \int_0^\infty x f(x) dx = \int_0^M x f(x) dx + \int_M^\infty x f(x) dx \\ &\geq \int_M^\infty x f(x) dx \geq M \int_M^\infty f(x) dx = M P\{X \geq M\} \end{aligned}$$

where f denotes the pdf of X . A similar argument works for other types of random variables (not continuous) as well. □

The third inequality (Chebyshev's Inequality) is a corollary of Markov's inequality. It provides an upper bound on the probability that a random variable falls a certain distance from its expectation.

Theorem 3 (Chebyshev's Inequality). *Suppose that X is a random variable such that $\sigma^2 = V(X) < \infty$. Then, for all $M > 0$,*

$$P(|X - \mathbb{E}[X]| \geq M) \leq \frac{\sigma^2}{M^2}.$$

Proof. Let $Y = (X - \mathbb{E}[X])^2$. Apply Markov's inequality to Y and the cutoff M^2 to get

$$P(Y \geq M^2) \leq \frac{\mathbb{E}[Y]}{M^2}.$$

Rewrite to get that

$$P(|X - \mathbb{E}[X]| \geq M) \leq \frac{\sigma^2}{M^2}.$$

□

Lastly, we state one more useful inequality involving random variables.

Theorem 4 (Holder's Inequality). *If $p > 1$ and $1/p + 1/q = 1$, and if $\mathbb{E}[|X|^p] < \infty$ and $\mathbb{E}[|Y|^q] < \infty$, then $\mathbb{E}[|XY|] \leq (\mathbb{E}[|X|^p])^{1/p} (\mathbb{E}[|Y|^q])^{1/q}$.*

1.1 Concentration Inequalities

Chebyshev's inequality delivers a finite-sample bound on the tail behavior of a sample average. Suppose $X_1, \dots, X_n$ are i.i.d. random variables with mean μ and variance $\sigma^2 < \infty$, and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Since $V(\bar{X}_n) = \sigma^2/n$, Chebyshev's inequality implies that, for any $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}.$$

This bound holds at every sample size n , but it shrinks only *polynomially* in n . Can we do better? For bounded random variables, the answer is yes: the tail probability in fact shrinks *exponentially* in n . Inequalities of this form are called *concentration inequalities*, and they are the workhorse of modern statistical learning theory. We state the most useful one. The key ingredient is a bound on the moment generating function of a bounded random variable.

Lemma 1 (Hoeffding's Lemma). *Suppose Z is a random variable with $\mathbb{E}[Z] = 0$ and $Z \in [a, b]$ almost surely. Then, for all $t > 0$,*

$$\mathbb{E}[\exp(tZ)] \leq \exp\left(\frac{t^2(b-a)^2}{8}\right).$$

We omit the proof, which follows from the convexity of the exponential function together with a second-order expansion; see, for example, Chapter 2 of Wainwright (2019). With Hoeffding's Lemma in hand, the bound on the tail probability of the sample average follows from Markov's

inequality applied to the exponentiated sum. This proof technique is known as the *Chernoff bound*, and it is worth knowing in its own right.

Theorem 5 (Hoeffding's Inequality). *Suppose $X_1, \dots, X_n$ are i.i.d. random variables with mean μ and $X_i \in [a, b]$ almost surely. Then, for all $\epsilon > 0$,*

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq 2 \exp\left(-\frac{2n\epsilon^2}{(b-a)^2}\right).$$

Proof. Consider first the upper tail. For any $t > 0$, the map $x \mapsto \exp(tx)$ is strictly increasing, and so

$$P(\bar{X}_n - \mu \geq \epsilon) = P\left(\exp\left(t \sum_{i=1}^n (X_i - \mu)\right) \geq \exp(tn\epsilon)\right) \leq \exp(-tn\epsilon) \mathbb{E}\left[\exp\left(t \sum_{i=1}^n (X_i - \mu)\right)\right],$$

where the inequality is Markov's inequality. Since the X_i are i.i.d., the expectation factors into the product $\prod_{i=1}^n \mathbb{E}[\exp(t(X_i - \mu))]$, and applying Hoeffding's Lemma to each term $Z_i = X_i - \mu \in [a - \mu, b - \mu]$ yields

$$P(\bar{X}_n - \mu \geq \epsilon) \leq \exp\left(-tn\epsilon + \frac{nt^2(b-a)^2}{8}\right).$$

This bound holds for every $t > 0$, and so we may choose t to minimize the right-hand side. Solving the first-order condition yields $t = \frac{4\epsilon}{(b-a)^2}$, and substituting this choice in gives

$$P(\bar{X}_n - \mu \geq \epsilon) \leq \exp\left(-\frac{2n\epsilon^2}{(b-a)^2}\right).$$

The same argument applied to $-X_i$ bounds the lower tail, and combining the two bounds delivers the result. $\square$

Compare Hoeffding's inequality with the Chebyshev bound above: for bounded random variables, the tail probability falls like $\exp(-cn\epsilon^2)$ rather than $\frac{1}{n\epsilon^2}$. Both bounds hold at every finite n . This is a different kind of statement than the asymptotic approximations we develop in the remainder of this lecture note: the law of large numbers and the central limit theorem describe the behavior of $\bar{X}_n$ as $n \rightarrow \infty$, whereas a concentration inequality is a guarantee at the sample size the researcher actually has.

Example 1. Suppose the researcher has trained a classifier and wishes to measure its accuracy on a held-out test set. For each test observation $i = 1, \dots, n$, let $X_i = 1$ if the classifier is correct and $X_i = 0$ otherwise, so $X_i \in [0, 1]$ and μ is the classifier's true accuracy. How large must the test set be so that the measured accuracy $\bar{X}_n$ lies within one percentage point of μ with probability at least 99%? Setting $2\exp(-2n\epsilon^2) \leq \delta$ and solving for n yields

$$n \geq \frac{\ln(2/\delta)}{2\epsilon^2},$$

and plugging in $\epsilon = 0.01$ and $\delta = 0.01$ gives $n \geq 26,492$. No asymptotic approximation is involved: the guarantee holds exactly at this sample size. Notice also what happens if the researcher instead compares M classifiers on the same test set. Applying Hoeffding's inequality to each classifier and taking a union bound, the probability that *any* measured accuracy misses by more than ϵ is at most $2M \exp(-2n\epsilon^2)$, and so it suffices that $n \geq \frac{\ln(2M/\delta)}{2\epsilon^2}$. The required sample size grows only logarithmically in the number of classifiers — it is the exponential tail that makes this possible, as a Chebyshev-based union bound would instead require n to grow linearly in M .

Remark 1. Hoeffding's inequality uses only the range of X_i , and it is therefore conservative when the variance is small relative to the range. Bernstein's inequality refines it: under the same conditions,

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq 2 \exp \left(-\frac{n\epsilon^2}{2\sigma^2 + \frac{2}{3}(b-a)\epsilon} \right),$$

which is sharper whenever $\sigma^2 \ll (b-a)^2$. There is a large family of results in this spirit — for sums of unbounded random variables with light tails, and for general functions of independent random variables that are insensitive to each coordinate (McDiarmid's inequality) — and they underpin the theory of modern machine learning methods. We will meet one more member of this family in the next lecture note: the Dvoretzky-Kiefer-Wolfowitz inequality, which is a concentration inequality for the empirical distribution function that holds *uniformly* over its argument.

2 Types of Convergence

Recall the definition of convergence for a non-stochastic sequence of real numbers. Let $\{x_n\}$ be a sequence of real numbers. We say

$$\lim_{n \rightarrow \infty} x_n = x$$

if for all $\epsilon > 0$, there exists some N such that for all $n > N$, $|x_n - x| < \epsilon$. We want to generalize this to the convergence of random variables. That is, under what conditions does the sequence of random variables $\{X_n\}$ “converge” to another random variable X ? There are several notions of stochastic convergence that are important to know.

Definition 1. The sequence of random variables $\{X_n\}$ *converges to the random variable X almost surely* if

$$P(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1.$$

We write $X_n \xrightarrow{a.s.} X$.

What does almost sure convergence mean? For a given outcome $\omega \in \Omega$, we can ask whether

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$$

holds using the definition of non-stochastic convergence. If the set of outcomes for which this holds has probability one then $X_n \xrightarrow{a.s.} X$.

Definition 2. The sequence of random variables $\{X_n\}$ *converges to the random variable X in probability* if, for all $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0.$$

We write $X_n \xrightarrow{p} X$.

What does convergence in probability mean? Fix an $\epsilon > 0$ and compute

$$P_n(\epsilon) = P(|X_n - X| > \epsilon).$$

This is just a number and so, we can check whether $P_n(\epsilon) \rightarrow 0$ using the definition of non-stochastic convergence. If $P_n(\epsilon) \rightarrow 0$ for all values $\epsilon > 0$, then $X_n \xrightarrow{p} X$.

Definition 3. The sequence of random variables $\{X_n\}$ *converges in mean to the random variable X* if

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|] = 0.$$

We write $X_n \xrightarrow{m} X$. We say $\{X_n\}$ *converges in mean-square to X* if

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|^2] = 0.$$

We write $X_n \xrightarrow{m.s.} X$.

Again, how do we interpret these notions of convergence? First, consider $m_n = \mathbb{E}[|X_n - X|]$, which is just a number. As a result, we say $X_n \xrightarrow{m} X$ if and only if $m_n \rightarrow 0$ using the definition of non-stochastic convergence. Similarly, $ms_n = \mathbb{E}[|X_n - X|^2]$ is also just a number and we can think about mean-square convergence in the same way.

Definition 4. Let $\{X_n\}$ be a sequence of random variables and $F_n(\cdot)$ is the cdf of X_n . Let X be a random variable with cdf $F(\cdot)$. $\{X_n\}$ **converges in distribution** or **weakly converges** to X if

$$\lim_{n \rightarrow \infty} F_n(x) = F(x)$$

for all points x at which $F(x)$ is continuous. We write $X_n \xrightarrow{d} X$.

Convergence in distribution describes the convergence of the cdfs. It does not mean that the realizations of the random variables will be close to each other. Recall that $F(x) = P(X \leq x) = P(\{\omega \in \Omega : X(\omega) \leq x\})$. As a result, $F_n(x) \rightarrow F(x)$ does not make any statement about $X_n(\omega)$ getting close to $X(\omega)$ for any $\omega \in \Omega$.

Why is convergence in distribution restricted to the continuity points of $F(x)$? An example may help. Let X_n be a degenerate random variable defined by $X_n = 1/n$ with probability 1 and let X be a degenerate random variable defined by $X = 0$ with probability one. Then, $F_n(x) = 1(x \geq 1/n)$ and $F(x) = 1(x \geq 0)$ with $F_n(0) = 0$ for all n while $F(0) = 1$. However, as $n \rightarrow \infty$, X_n is

getting closer and closer to X in the sense that for all $x \neq 0$, $F_n(x)$ is well approximated by $F(x)$. Alternatively, if we did not restrict convergence in distribution to the continuity points, we would have the strange case where a non-stochastic sequence $\{X_n\}$ converges to X under the non-stochastic definition of convergence but does not converge in distribution.

Multivariate Extension: We can extend each of these definitions to random vectors. For example, the sequence of random vectors $\{X_n\} \xrightarrow{a.s.} X$ if each element of X_n converges almost surely to each element of X . The extension is analogous for convergence in probability.

A sequence of random vectors converges in distribution to a random vector if we apply the definition above to the joint cumulative distribution function. Alternatively, the following theorem provides another characterization of multivariate convergence in distribution.

Theorem 6 (Cramer-Wold Device). *Let $\{Z_n\}$ be a sequence of k -dimensional random vectors. Then, $Z_n \xrightarrow{d} Z$ if and only if $\lambda'Z_n \xrightarrow{d} \lambda'Z$ for all $\lambda \in \mathbb{R}^k$.*

2.1 Relationships between Types of Convergence

How do these different definitions of stochastic convergence relate to each other? The next set of propositions lay out the relationships. The key relationships are summarized in Figure 1.

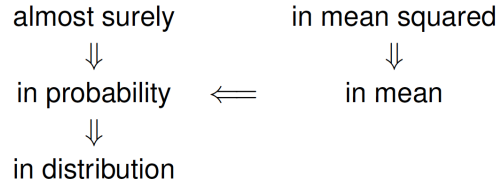


Figure 1: Relationships between Types of Convergence

Theorem 7.

1. Convergence in mean-square implies convergence in mean: Suppose $X_n \xrightarrow{ms} X$. Then, $X_n \xrightarrow{m} X$.
2. Convergence in mean-square implies convergence in probability: Suppose $X_n \xrightarrow{ms} X$. Then, $X_n \xrightarrow{p} X$.
3. Convergence in mean implies convergence in probability: Suppose $X_n \xrightarrow{m} X$. Then, $X_n \xrightarrow{p} X$.
4. Almost sure convergence implies convergence in probability: Suppose $X_n \xrightarrow{a.s.} X$. Then, $X_n \xrightarrow{p} X$.

5. Convergence in probability implies convergence in distribution: Suppose $X_n \xrightarrow{p} X$. Then, $X_n \xrightarrow{d} X$.

Proof. (1) follows from Jensen's Inequality. Recall that if $h(\cdot)$ is a convex function, then

$$\mathbb{E}[h(Y)] \geq h(\mathbb{E}[Y]).$$

Set $h(z) = z^2$ and $Y = |X_n - X|$. It follows that

$$0 \leq \mathbb{E}[|X_n - X|]^2 \leq \mathbb{E}[|X_n - X|^2]$$

with $\mathbb{E}[|X_n - X|^2] \rightarrow 0$. The result follows.

(2) follows from Markov's Inequality. Recall that, if Y is non-negative, then, for all $c > 0$,

$$P(Y \geq c) \leq \mathbb{E}[Y]/c.$$

Fix $\epsilon > 0$. Set $c = \epsilon^2$ and $Y = |X_n - X|^2$. We have that

$$0 \leq P(|X_n - X|^2 \geq \epsilon^2) = P(|X_n - X| \geq \epsilon) \leq \mathbb{E}[|X_n - X|^2]/\epsilon^2.$$

Taking limits of both sides, we get that

$$0 \leq \lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|^2]/\epsilon^2 = 0$$

and the result follows.

(3) also follows from Markov's Inequality by a similar argument to (2). (4) is left as an exercise.

To prove (5), we note that $X_n \leq x$ and $X > x + \epsilon$ implies $|X_n - X| > \epsilon$. Thus,

$$\begin{aligned} F_{X_n}(x) &= P\{X_n \leq x\} \\ &= P\{X_n \leq x, X \leq x + \epsilon\} + P\{X_n \leq x, X > x + \epsilon\} \\ &\leq P\{X \leq x + \epsilon\} + P\{|X_n - X| > \epsilon\} \\ &= F_X(x + \epsilon) + P\{|X_n - X| > \epsilon\}. \end{aligned}$$

for any $x \in \mathbb{R}$ and $\epsilon > 0$. Similarly,

$$F_X(x - \epsilon) \leq F_{X_n}(x) + P\{|X_n - X| > \epsilon\}.$$

Thus,

$$F_X(x - \epsilon) - P\{|X_n - X| > \epsilon\} \leq F_{X_n}(x) \leq F_X(x + \epsilon) + P\{|X_n - X| > \epsilon\}.$$

Next, if x is a point of continuity of F_X , for any $\delta > 0$, there exists $\epsilon(\delta) > 0$ such that

$$F_X(x + \epsilon(\delta)) - \delta \leq F_X(x) \leq F_X(x - \epsilon(\delta)) + \delta.$$

Therefore,

$$F_X(x) - \delta - P\{|X_n - X| > \varepsilon(\delta)\} \leq F_{X_n}(x) \leq F_X(x) + \delta + P\{|X_n - X| > \varepsilon(\delta)\}.$$

Next, since $X_n \rightarrow_p X$, by definition it follows that:

$$\lim_n |F_{X_n}(x) - F_X(x)| \leq \delta$$

So, $F_{X_n}(x) \rightarrow F_X(x)$ as $n \rightarrow \infty$ for any $x \in \mathbb{R}$ where $F_X(x)$ is continuous. □

2.2 Slutsky Theorem and the Continuous Mapping Theorem

We conclude this section with two theorems that are very useful in deriving asymptotic distributions.

Theorem 8 (Slutsky Theorem). *Let c be a constant. Suppose that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$. Then,*

1. $X_n + Y_n \xrightarrow{d} X + c$.
2. $X_n Y_n \xrightarrow{d} Xc$.
3. $X_n/Y_n \xrightarrow{d} X/c$ provided that $c \neq 0$.

If $c = 0$, then $X_n Y_n \xrightarrow{p} 0$.

Theorem 9 (Continuous Mapping Theorem). *Let $g(\cdot)$ be a continuous function. Then,*

1. *If $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$.*
2. *If $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.*

Proof. We provide the proof for (2) and the case where $X = a \in \mathbb{R}$. Let $\epsilon > 0$. Since $g(\cdot)$ is continuous at a , there exists some $\delta > 0$ such that

$$|x - a| < \delta \implies |g(x) - g(a)| < \epsilon.$$

The contrapositive of this is

$$|g(x) - g(a)| \geq \epsilon \implies |x - a| \geq \delta.$$

Substituting in X_n , it follows that

$$P(|g(X_n) - g(a)| \geq \epsilon) \leq P(|X_n - a| \geq \delta) \rightarrow 0.$$

The result follows. □

3 O_p and o_p Notation

First, let us talk about some notation for *non-stochastic* sequences x_n and b_n . We would write $x_n = o(b_n)$ when $\lim_{n \rightarrow \infty} \frac{x_n}{b_n} = 0$. Often it is described as “ x_n is asymptotically smaller than b_n ”. We would write $x_n = O(b_n)$ when $\sup_n \left| \frac{x_n}{b_n} \right| < \infty$. In general we will use some easy-to-describe sequences as b_n , such as $\sqrt{n}$, $\frac{1}{n}$ or 1. Notice that $x_n = o(1)$ means $x_n \rightarrow 0$ and $x_n = O(1)$ means that x_n is a bounded sequence.

Now let us adopt similar notations for sequences of random variables $\{X_n\}_{n=1}^{\infty}$ (in the same spirit as how we generalized our notion of convergence for non-stochastic sequences earlier).

Definition 5.

- We say that $X_n = o_p(b_n)$ if and only if $\frac{X_n}{b_n} \rightarrow_p 0$.
- We say that $X_n = O_p(b_n)$ if and only if for any $\varepsilon > 0$ there exists a constant $C < \infty$ such that $P\left\{\left|\frac{X_n}{b_n}\right| > C\right\} < \varepsilon$ for all n .

Sometimes statement $X_n = O_p(b_n)$ is described as “the sequence $\frac{X_n}{b_n}$ is stochastically bounded.”

Some examples of the use of these symbols are given below:

- If $X_n \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$ then $X_n = O_p(1)$;
- If $X_n \rightarrow_p X$ then $X_n = X + o_p(1)$;
- Chebyshev’s inequality for i.i.d. X_i with mean μ and finite variance implies $\bar{X}_n - \mu = O_p(\frac{1}{\sqrt{n}})$

Remark 2. The first fact in the list above generalizes: if $X_n \xrightarrow{d} X$ for any random variable X , then $X_n = O_p(1)$. The converse is *false*: $X_n = O_p(1)$ does not imply that X_n converges in distribution. For a counterexample, let $X_n \sim U[-1, 0]$ if n is odd and $X_n \sim U[0, 1]$ if n is even. For any $\varepsilon > 0$, choose any $C > 1$. Then $P(|X_n| > C) = 0 < \varepsilon$ for all n , so $X_n = O_p(1)$. However, for any $x \in (-1, 0) \cup (0, 1)$, the cdfs $F_n(x)$ oscillate between zero and a strictly positive number as n alternates between odd and even, so $F_n(x)$ does not converge and $\{X_n\}$ does not converge in distribution. Stochastic boundedness controls the tails of the X_n uniformly in n ; it does not require the distributions of the X_n to settle down to any limit.

We will often use the following statements:

- If $X_n = O_p(n^{-\delta})$ for some $\delta > 0$ then $X_n = o_p(1)$;
- If $X_n = o_p(b_n)$ then $X_n = O_p(b_n)$;
- If $X_n = O_p(n^\alpha)$ and $Y_n = O_p(n^\beta)$, then $X_n Y_n = O_p(n^{\alpha+\beta})$ and $X_n + Y_n = O_p(\max\{n^\alpha, n^\beta\})$;
- If $X_n = O_p(n^\alpha)$ and $Y_n = o_p(n^\beta)$, then $X_n Y_n = o_p(n^{\alpha+\beta})$;
- If $X_n = O_p(n^\alpha)$ and $Y_n = o_p(n^\alpha)$, then $X_n + Y_n = O_p(n^\alpha)$

Example 2. Let $X_n \sim N(0, n)$. Then,

$$X_n = O_p(n^{1/2}).$$

Why? We have that $X_n/n^{1/2} \sim N(0, 1)$ for all n . For any $\epsilon > 0$, we can choose an M such that $P(|N(0, 1)| < M) > 1 - \epsilon$. We also have that

$$X_n = o_p(n).$$

Why? We have that $X_n/n \sim N(0, 1/n)$. Note that

$$P(|N(0, 1/n)| > \epsilon) = P(|N(0, 1)| > n^{1/2}\epsilon) \rightarrow 0.$$

Alternatively, note that

$$\mathbb{E}[(X_n/n - 0)^2] = V(X_n/n) = 1/n \rightarrow 0$$

and so, $X_n \xrightarrow{ms} 0$.

Example 3. How do we exhibit the constant C in the definition of O_p in practice? Chebyshev's inequality provides a general recipe with an explicit constant. Let $\{Y_n\}$ be a sequence of random variables with $\mathbb{E}[Y_n]$ finite and $V(Y_n) < \infty$ for each n . We claim that

$$Y_n - \mathbb{E}[Y_n] = O_p\left(\sqrt{V(Y_n)}\right).$$

Fix $\epsilon > 0$. By Chebyshev's inequality, for any $M > 0$,

$$P(|Y_n - \mathbb{E}[Y_n]| \geq M) \leq \frac{V(Y_n)}{M^2}.$$

Choose the constant $C = \sqrt{2/\epsilon}$ and set $M = \sqrt{V(Y_n)}C$. Then, for all n ,

$$P\left(\left|\frac{Y_n - \mathbb{E}[Y_n]}{\sqrt{V(Y_n)}}\right| > C\right) \leq P(|Y_n - \mathbb{E}[Y_n]| \geq \sqrt{V(Y_n)}C) \leq \frac{V(Y_n)}{V(Y_n)C^2} = \frac{\epsilon}{2} < \epsilon.$$

The single constant $C = \sqrt{2/\epsilon}$ works for every n , which is exactly what the definition of O_p requires. In particular, if $X_1, X_2, \dots$ are i.i.d. with mean μ and variance σ^2 , applying this argument to $Y_n = \bar{X}_n$ delivers $\bar{X}_n - \mu = O_p(1/\sqrt{n})$ since $V(\bar{X}_n) = \sigma^2/n$, verifying the third fact in the first bulleted list above.

4 Law(s) of Large Numbers

There are two basic building blocks that we use to construct asymptotic results. We introduce the first building block in this section: laws of large numbers (LLNs). These results establish that sample averages converge to expectations. There are many LLNs under alternative assumptions on the data-generating process (for example, for time series settings). We focus on two key LLNs.

Theorem 10 (Chebyshev's Weak Law of Large Numbers). *Let $X_1, X_2, \dots$ be a sequence of random variables with $\mathbb{E}[X_i] = \mu_i$, $V(X_i) = \sigma_i^2$ and $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$. Define*

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \bar{\mu}_n = \frac{1}{n} \sum_{i=1}^n \mu_i, \quad \bar{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n \sigma_i^2$$

and assume that $\bar{\sigma}_n^2/n \rightarrow 0$. Then,

$$\bar{X}_n - \bar{\mu}_n \xrightarrow{p} 0.$$

Proof. Observe that

$$\mathbb{E}[\bar{X}_n - \bar{\mu}_n] = 0.$$

Further observe that

$$\begin{aligned} V(\bar{X}_n - \bar{\mu}_n) &= V(\bar{X}_n) \\ &= \frac{1}{n^2} \sum_{i,j} \text{Cov}(X_i, X_j) \\ &= \frac{1}{n^2} \sum_i \sigma_i^2 = \bar{\sigma}_n^2/n \rightarrow 0. \end{aligned}$$

Therefore, $\bar{X}_n - \bar{\mu}_n \xrightarrow{ms} 0$ and so, $\bar{X}_n - \bar{\mu}_n \xrightarrow{p} 0$. □

Theorem 11 (Strong law of large numbers). *If $X_1, X_2, \dots$ are i.i.d. with $\mathbb{E}[|X_i|] < \infty$ and $\mathbb{E}[X_i] = \mu$, then*

$$\bar{X}_n \xrightarrow{as} \mu.$$

Notice that for the weak law of large numbers, we only required the sequence of X_i 's to be uncorrelated and we required finite second moments. For the strong law of large numbers, we required that the X_i 's are i.i.d., but we did not have to introduce assumptions about the second moments.

5 Central Limit Theorem(s)

The second building block for asymptotic results is the Central Limit Theorem (CLT). CLTs show that properly centered sample averages will converge in distribution to normal random variables.

Theorem 12 (Central limit theorem). *Let $\{X_i\}$ be a sequence of i.i.d. random variables with mean μ and variance σ^2 . Then*

$$\sum_{i=1}^n (X_i - \mu)/\sqrt{n} \xrightarrow{d} N(0, \sigma^2).$$

In the multivariate case, if $\text{Var}(X_i) = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_i - \mathbb{E}[X_i])^T] = \Sigma$, then $\sum_{i=1}^n (X_i - \mu)/\sqrt{n} \xrightarrow{d} N(0, \Sigma)$.

We often will need to consider non-identically distributed random variables; in such a case we should use the Lindeberg–Feller CLT:

Theorem 13 (Lindeberg–Feller). *Let $\{X_i\}_{i=1}^\infty$ be a sequence of independent random variables with $\mathbb{E}X_i = \mu_i$ and $V(X_i) = \sigma_i^2$. Denote $c_n^2 = V(\sum_{i=1}^n X_i) = \sum_{i=1}^n \sigma_i^2$. If for any $\varepsilon > 0$*

$$\lim_{n \rightarrow \infty} \frac{1}{c_n^2} \sum_{i=1}^n \mathbb{E} \left((X_i - \mu_i)^2 I_{\{|X_i - \mu_i| > \varepsilon c_n\}} \right) = 0 \quad (1)$$

then $\frac{\sum_{i=1}^n (X_i - \mu_i)}{c_n} \xrightarrow{d} N(0, 1)$.

Sometimes Lindeberg’s condition (1) is called “asymptotic negligibility” as in particular it implies that $\max_{1 \leq i \leq n} \frac{\sigma_i^2}{c_n^2} \rightarrow 0$ and guarantees that the contribution of any individual X_i to the variance of the sum is sufficiently small for large n . The following sufficient condition for (1) is called Lyapunov’s:

$$\lim_{n \rightarrow \infty} \frac{1}{c_n^{2+\delta}} \sum_{i=1}^n \mathbb{E} |X_i - \mu_i|^{2+\delta} = 0 \text{ for some } \delta > 0.$$

6 The Delta Method

Theorem 14. *Assume that for a sequence of random variables X_n and constants μ and σ we have $\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2)$, and let $g(\cdot)$ be continuously differentiable. If $g'(\mu) \neq 0$, then*

$$\sqrt{n}(g(X_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2 (g'(\mu))^2).$$

Proof. By the mean value theorem, for any realization $X_n(\omega)$, there is some $\mu_n^*(\omega)$ between μ and $X_n(\omega)$ such that

$$g(X_n(\omega)) - g(\mu) = g'(\mu_n^*)(X_n(\omega) - \mu). \quad (2)$$

Thus, we have defined a new sequence of random variables, $\{\mu_n^*\}_{n=1}^\infty$. By assumption, we have $\sqrt{n}(X_n - \mu) = O_p(1)$, thus $(X_n - \mu) = o_p(1)$ and $X_n \xrightarrow{p} \mu$. Since μ_n^* is between μ and X_n , $\mu_n^* \xrightarrow{p} \mu$ as well. By the Continuous mapping theorem, $g'(\mu_n^*) \rightarrow_p g'(\mu)$ since $g'(x)$ is continuous. Moreover, by the Slutsky theorem and by the Central limit theorem, $\sqrt{n}g'(\mu_n^*)(X_n - \mu) \xrightarrow{d} g'(\mu)N(0, \sigma^2)$. $\square$

The following alternative proof replaces the mean value theorem with a Taylor expansion carrying an explicit o_p remainder. It illustrates the O_p/o_p algebra from Section 3 in action.

Alternative proof via O_p/o_p algebra. Define

$$h(x) = \frac{g(x) - g(\mu)}{x - \mu} - g'(\mu) \text{ for } x \neq \mu, \quad h(\mu) = 0.$$

Differentiability of $g(\cdot)$ at μ means exactly that $h(x) \rightarrow 0$ as $x \rightarrow \mu$, so $h(\cdot)$ is continuous at μ . Rearranging the definition of $h(\cdot)$, we may write the Taylor expansion

$$g(X_n) = g(\mu) + g'(\mu)(X_n - \mu) + h(X_n)(X_n - \mu).$$

Multiplying both sides by $\sqrt{n}$ and rearranging,

$$\sqrt{n}(g(X_n) - g(\mu)) = g'(\mu)\sqrt{n}(X_n - \mu) + h(X_n)\sqrt{n}(X_n - \mu).$$

By assumption, $\sqrt{n}(X_n - \mu) = O_p(1)$, and therefore $X_n - \mu = O_p(n^{-1/2}) = o_p(1)$; that is, $X_n \xrightarrow{p} \mu$. Since $h(\cdot)$ is continuous at μ with $h(\mu) = 0$, the Continuous Mapping Theorem gives $h(X_n) \xrightarrow{p} 0$, i.e. $h(X_n) = o_p(1)$. Applying the O_p/o_p algebra from Section 3, the remainder term satisfies

$$h(X_n)\sqrt{n}(X_n - \mu) = o_p(1)O_p(1) = o_p(1).$$

By the Continuous Mapping Theorem, $g'(\mu)\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2)$, and by the Slutsky theorem adding the $o_p(1)$ remainder does not change the limiting distribution. Therefore

$$\sqrt{n}(g(X_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2).$$

Notice that this argument replaces the random intermediate point μ_n^* from the mean value theorem with an explicit $o_p(1)$ remainder, and only requires $g(\cdot)$ to be differentiable at μ . $\square$

Note that this theorem also holds when $g'(\mu) = 0$ but in this case the asymptotic distribution will be 0 (constant), i.e. degenerate. This style of argument is very common in statistics and econometrics.

The Delta method has a multidimensional extension. Let $X_1, \dots, X_n, \dots$ be a sequence of iid $k \times 1$ random vectors with mean μ and covariance matrix Σ . Then, by the multidimensional Central limit theorem, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \Sigma)$. Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$ be a twice continuously differentiable function. Let $\tau^2 = (\partial g(\mu)/\partial \mu)^T \Sigma (\partial g(\mu)/\partial \mu)$. Here $\partial g(\mu)/\partial \mu$ is a $k \times 1$ vector with i -th component equal to $\partial g(\mu)/\partial \mu_i$. Then,

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N(0, \tau^2).$$

Example 4. Let $X_1, \dots, X_n, \dots$ be a sequence of iid random variables with mean μ and variance σ^2 . What is the limiting distribution of $(\bar{X}_n)^2$? Let $g(x) = x^2$. Then $g'(\mu) = 2\mu$. Thus, by the Delta method, $\sqrt{n}((\bar{X}_n)^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$. Note that if $\mu = 0$, then the limit distribution is degenerate.